Adify Skills
n8n Automation Mastery

📘 CLASS NOTES

🔌 Module 3 · APIs, Webhooks & Integrations

DAY 08 of 25

Authentication, Credentials & Webhooks

“Real-time apps banane ka secret: webhooks.”

⚡ AAJ HUM YEH BANAYENGE

Website Form → Webhook Lead API

Apna mini API: website form POST karta hai, n8n validate karta hai, Sheet mein save karta hai, customer ko auto-reply bhejta hai aur website ko JSON response deta hai (200 ya 400).

Webhook | Respond to Webhook | OAuth2

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Webhooks se n8n ek "backend" ban jaata hai — website, app, payment gateway sab real-time mein n8n ko data bhej sakte hain.

🎯 Aaj ke baad aap yeh kar paoge

01 API keys, Bearer tokens, Basic auth

🌍 Gym card (API key), movie ticket (bearer token), ghar ki chaabi (basic auth: user+password).
⚙️ HTTP Request → Authentication: Header Auth → x-api-key: ••••

02 OAuth2 flow — Google, Microsoft, social apps

🌍 Valet parking key — gaadi chala sakta hai, dikki nahi khol sakta.
⚙️ Google OAuth2 credential → sirf Sheets scope.

03 n8n credential manager & sharing

🌍 Ghar ki chaabiyon ka ek locked box — family ko share.
⚙️ Credentials → New → ek baar banao, har workflow mein select.

04 Webhook node: test vs production URLs

🌍 Rehearsal ka stage vs asli show ka stage.
⚙️ /webhook-test/lead (sirf Listen mode) vs /webhook/lead (active).

05 Respond to Webhook — build your own API

🌍 Courier ko receipt dena.
⚙️ Respond to Webhook: 200 { "ok": true, "id": "lead_12" }

06 Webhook security: auth headers, HMAC verify

🌍 Note par watermark — asli hai ya nakli.
⚙️ Crypto node: HMAC-SHA256(body, secret) === header signature?

⏱ Class ka plan (2 ghante)

10m	35m	45m	20m	10m

0:00-0:10 Recap

🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

0:10-0:45 Concept: Auth & Webhooks

Push vs pull

🪄 Tumhare ghar ka pata (webhook) sabko pata hai, par andar sirf woh aa sakta hai jiske paas pass (auth) hai.

0:45-1:30 Live Build

Apna mini REST API: /lead endpoint with validation

🪄 Aaj ki kahani yaad hai — 'Society gate aur security guard'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

1:30-1:50 Challenge

Secure the webhook with header auth

🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.

1:50-2:00 Q&A + Homework

🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Webhook URL

Ek unique address jahan data POST hota hai.

🪄 **Analogy:** Aapka ghar ka pata — courier wahin aata hai

CONCEPT 02

Test vs Production URL

Test URL sirf "Listen" mode mein; production URL active workflow par.

🪄 **Analogy:** Rehearsal vs real show

CONCEPT 03

Respond to Webhook

Caller ko apna custom jawab bhejo — n8n ek API ban gaya.

🪄 **Analogy:** Courier ko signature dena

CONCEPT 04

OAuth2

Bina password diye app ko permission dena (Google, Microsoft).

🪄 **Analogy:** Valet parking key — sirf gaadi chalane ki permission

🔑 Aaj ka main idea

Authentication = "aap kaun ho" proof (API key, token, OAuth). Webhook = n8n ka public pata jahan apps data bhejte hain. Respond to Webhook se n8n wapas jawab deta hai — yaani n8n khud ek API ban jaata hai.

- API key = gym membership card
- OAuth = "Google se login karo" button
- Webhook = Swiggy restaurant ko order notification

📖 Ek kahani se samjho: Society gate aur security guard

1 🏠 **Ghar ka pata diya** = Webhook URL

2 💂 **Guard ID check karta hai** = Header Auth / HMAC

3 🗝️ **Valet ko sirf gaadi ki chaabi** = OAuth2 (limited permission)

4 📋 **Visitor register** = Google Sheets

5 ✅ **Guard bola "andar aao"** = Respond to Webhook

🪄 Ek line mein: “Tumhare ghar ka pata (webhook) sabko pata hai, par andar sirf woh aa sakta hai jiske paas pass (auth) hai.”

🗺️ Aaj ka workflow: Website Form → Webhook Lead API

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

Website owners, agencies (form → API → CRM) — aam taur par **₹8,000 – ₹20,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1 ⚡ **Webhook**
POST /lead, Authentication: Header Auth (x-api-key)

2 🔀 **IF**
name, phone, email present hain?

3 📊 **Google Sheets**
Valid lead append

4 📧 **Gmail**
Lead ko auto-reply

5 ⚡ **Respond to Webhook**
200 { ok: true, id } ya 400 { error }

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Connect: **Google Sheets (Google OAuth)**
- ☐ Connect: **Gmail (Google OAuth)**
- ☐ Sheet mein "Website Leads" tab banao
- ☐ Production mein Webhook par Header Auth lagao (Authentication → Header Auth)

✅ Test kaise karein

1. Webhook → "Listen for test event"
2. Neeche curl chalao — valid aur invalid dono body se
3. Response code 200/400 check karo

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😩 Pehle (manual)

Website form email mein aata	5 min
Manually Sheet mein	5 min
Customer ko reply	5 min
	15 min

➜

⚡ Ab (n8n)

Webhook → validate → Sheet → auto-reply → response

0 min

⏱ Har submission par 15 min bache, apna mini API ready.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
curl -X POST https://your-n8n.com/webhook/lead \
  -H "x-api-key: SECRET" -H "Content-Type: application/json" \
  -d '{"name":"Priya","phone":"9123456780"}'
```

📦 Sample data — input se output tak

📥 Input — 📋 Select karke copy karo

```
{
  "name": "Priya",
  "phone": "9123456780",
  "email": "priya@mail.com"
}
```

📤 Output — 📋 Select karke copy karo

```
{
  "ok": true,
  "id": "lead_1043"
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude — 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Apna mini API: website form POST karta hai, n8n validate karta hai, Sheet mein save karta hai,
customer ko auto-reply bhejta hai aur website ko JSON response deta hai (200 ya 400).

Flow: Webhook (POST /lead) → IF (Name + valid email?) → Google Sheets (Save) → Gmail (Auto-reply) →
Respond to Webhook (200 / 400 JSON)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to
test it and the 3 most common errors beginners hit.
```

💬 **Debug helper prompt** 📋 Select karke copy karo

```
I'm building the n8n workflow "Website Form → Webhook Lead API" (nodes: POST /lead, Valid?, Save Lead,
Auto Reply, Respond 200, Respond 400).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **QR scan karo** ya link kholo — workflow ki JSON file khulegi.
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
6 nodes · ready to import
adifyskills.com/workflows/day-08-webhook-lead-api.json

🌐 **Lesson page**
Copy buttons, quiz aur video links
adifyskills.com/day/8

Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Gmail (Google OAuth)" connect karo.
Webhook/Form test par kuch nahi ho raha?	"Listen for test event" / Test URL use karo aur 2 minute ke andar request bhejo. Production URL sirf Active workflow par chalta hai.
"Sheet not found" / column khaali?	Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye.

⚠️ Yeh galtiyan sabse common hain

- ! Test URL website par laga dena (workflow band hote hi fail)
- ! Respond to Webhook add kiya par Webhook ka "Respond" setting change nahi kiya
- ! Webhook ko bina auth public chhod dena

⭐ Pro tips — time bachane wali tricks

- ★ Postman / Hoppscotch se webhook test karo
- ★ HMAC signature verify ke liye Crypto node
- ★ Webhook path readable rakho: /lead, /payment-success

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋 Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Website form → n8n Webhook → validation → Sheet + auto-reply → JSON response.

📝 Homework

Google OAuth2 credential setup karo aur ek private Google Sheet read karo.

- ☐ Homework ho gaya

BEGINNER
Webhook se data Sheet mein ☐

INTERMEDIATE
Validation errors ki list return karo ☐

ADVANCED
Razorpay-style HMAC signature verify ☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Webhook ko secure karo: bina sahi header ke request aaye to reject ho.

💡 Hint 1
Webhook node ki settings mein "Authentication" option hai.

💡 Hint 2
Header Auth chuno, credential mein Name: x-api-key, Value: koi secret.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Website par kaunsa URL lagana chahiye?

- (A) Test URL
- (B) Production URL
- (C) Koi bhi
- (D) Editor URL

Q2. Caller ko custom JSON dene ka node?

- (A) Code
- (B) HTTP Request
- (C) Respond to Webhook
- (D) Set

Q3. Google login ke liye auth type?

(A) None
(B) Header
(C) Basic
(D) OAuth2

Q4. Webhook secure karne ka simple tareeka?

(A) Long path
(B) Kuch nahi
(C) Capital letters
(D) Header Auth

📖 Aaj ke naye shabd

Workflow
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

Node
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana…).

Webhook
Ek URL jis par koi doosra app data bhejta hai aur aapka workflow turant chal padta hai.

API
Do apps ke baat karne ka tareeka. Ek app sawaal bhejta hai, doosra jawab (data) deta hai.

Credential
Kisi app ka login/API key jo n8n safe tareeke se yaad rakhta hai.

OAuth
Login ka safe tareeka — "Sign in with Google" jaisa, password share kiye bina.

✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 9

Google Workspace & Messaging Automation

"Gmail, Sheets, Drive, Calendar + WhatsApp, Telegram, Slack."

☐ **Day 8 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → B
Production URL

Q2 → C
Respond to Webhook

Q3 → D
OAuth2

Q4 → D
Header Auth

✅ Challenge ka solution

Webhook → Authentication: Header Auth → credential (x-api-key = mySecret123). curl mein -H "x-api-key: mySecret123" bhejo.

📚 Official docs

- Credentials ↗
- Webhook node ↗
- Respond to Webhook ↗
- Google OAuth setup ↗
- HTTP Request credentials ↗

▶ Videos

- n8n webhook tutorial ↗
- n8n respond to webhook api ↗

💬 Doubt poochho

WhatsApp par seedha message

`wa.me/919000000000?text=Day%208%20(Authentication%2C%20Credentials%20%26%20Webhooks)%20mein%20doubt%20hai%3A%20`

🌐 Day 8 lesson page

Quiz, copy buttons, videos

`adifyskills.com/day/8`